

Assignment # 2

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SQL:

1

```
SELECT TEAMNAME, CITY  
FROM TEAM  
WHERE TEAMNUM > 15  
ORDER BY TEAMNAME;
```

RA:

π TEAMNAME, CITY (σ TEAMNUM > 15 (TEAM))

2

SQL:

```
SELECT PLAYERNAME  
FROM PLAYER  
WHERE AGE < 18;
```

RA:

π PLAYERNAME (σ AGE < 18 (PLAYER))

3

SQL:

```
SELECT COACHNAME  
FROM COACH, WORKEXPERIENCE  
WHERE COACH.TEAMNUM = WORKEXPERIENCE.TEAMNUM  
AND COACHNAME LIKE 'R.%' AND  
EXPERIENCE TYPE = 'College' AND  
YEARS EXPERIENCE BETWEEN 5 AND 10;
```

RA:

$\pi_{\text{COACHNAME}} ((\text{COACH} \bowtie \text{COACH.TEAMNUM} = \text{WORKEXP} \text{ EXPERIENCE.TEAMNUM}) \bowtie (\text{COACHNAME LIKE 'R.I.' AND EXPERIENCE TYPE = 'College' AND YEARSEXPERIENCE BETWEEN 5 AND 10}) (\text{WORK EXPERIENCE}))$

4 SQL:

```
SELECT SUM (YEARSEXPERIENCE)
FROM WORK EXPERIENCE
WHERE TEAMNUM = 25 AND COACHNAME
= 'Waqar';
```

RA:

$\pi_{\text{SUM (YEARSEXPERIENCE)}} (\sigma_{\text{TEAMNUM} = 25 \text{ AND COACHNAME} = \text{'Waqar'}} (\text{WORK EXPERIENCE}))$

5 SQL:

```
SELECT COUNT (DISTINCT EXPERIENCE TYPE)
FROM WORK EXPERIENCE
WHERE TEAMNUM = 3 AND COACHNAME
= 'Waqar';
```

RA: $\pi_{\text{COUNT (DISTINCT EXPERIENCE TYPE)}} (\sigma_{\text{TEAMNUM} = 3 \text{ AND COACHNAME} = \text{'Waqar'}} (\text{WORK EXPERIENCE}))$

6

SQL:

```
SELECT COACHNAME, SUM(YEARSEXPERIENCE)
FROM WORK EXPERIENCE
WHERE TEAMNUM = 3
GROUP BY COACHNAME;
```

RA:

```
 $\pi_{COACHNAME, SUM(YEARSEXPERIENCE)}$ 
( $\sigma_{TEAMNUM = 3} (WORK EXPERIENCE)$ 
 $\bowtie COACHNAME (WORK EXPERIENCE)$ )
```

7

SQL:

```
SELECT COUNT(DISTINCT MANUFACTURER)
FROM BATS;
```

RA:

```
 $\pi_{COUNT(DISTINCT MANUFACTURER)}$  (BATS)
```

8

SQL:

```
SELECT PLAYERNAME
FROM PLAYER, AFFILIATION, TEAM
WHERE PLAYER.PLAYERNUM =
AFFILIATION.PLAYERNUM AND
AFFILIATION.TEAMNUM = TEAM.
TEAMNUM AND TEAM.TEAMNAME =
'Yankees' AND YEARS >= 5;
```


RA:

π PLAYERNAME ((PLAYER $\bowtie$ PLAYER,
PLAYERNUM = AFFILIATION . PLAYERNUM
AFFILIATION) $\bowtie$ (TEAM $\bowtie$ TEAM, TEAMNUM
= AFFILIATION . TEAMNUM & TEAMNAME
= 'Yankees') (AFFILIATION) & YEARS >= 5
(AFFILIATION))

9

SQL:

```
SELECT COACHNAME, SUM (YEARSEXPERIENCE)
FROM WORK EXPERIENCE, TEAM
WHERE WORK EXPERIENCE . TEAMNUM AND
= TEAM . TEAMNUM AND TEAM . TEAMNAME
= 'Royals' AND YEARSEXPERIENCE > 8
GROUP BY COACHNAME;
```

RA:

π COACHNAME, SUM (YEARSEXPERIENCE)
((WORK EXPERIENCE $\bowtie$ WORK EXPERIENCE,
TEAMNUM = TEAM . TEAMNUM & TEAMNAME
= 'Royals') (TEAM) & YEARSEXPERIENCE
> 8 (WORK EXPERIENCE) $\bowtie$ COACHNAME
(WORK EXPERIENCE))

10

SQL:

```
SELECT PLAYERNAME
FROM PLAYER
WHERE AGE = (SELECT MIN(AGE) FROM
              PLAYER);
```

RA: π PLAYERNAME (& ^{AGE} ~~AGE~~ = MIN (AGE)
(PLAYER))
(PLAYER))

11

SQL:

```
SELECT PLAYERNAME  
FROM PLAYER  
WHERE SUBSTRING (PLAYERNAME, -1)  
= 'M' AND AGE = (SELECT MIN(AGE)  
FROM PLAYER WHERE SUBSTRING  
(PLAYERNAME, -1) = 'M')
```

RA:

Youngest-M-players = σ SUBSTRING
(PLAYERNAME, -1) = 'M' AND AGE =
(π MIN(AGE) (σ SUBSTRING (PLAYERNAME, -1)
= 'M')) (PLAYER) (PLAYER)

12

SQL:

```
SELECT PLAYERNAME  
FROM PLAYER  
WHERE PLAYERNUM IN (SELECT PLAYERNUM  
FROM AFFILIATION  
GROUP BY PLAYERNUM  
HAVING COUNT (DISTINCT TEAMNUM) > 2)
```

RA: player-affiliated-with-multiple-teams

= π PLAYERNAME ((π PLAYERNUM
(AFFILIATION) $\div$ GROUP BY PLAYERNUM
(π TEAMNUM (AFFILIATIONS)))) $\div$ GROUP BY
PLAYERNUM ((π PLAYERNUM (AFFILIATION))
 $\div$ GROUP BY PLAYERNUM (π TEAMNUM
(AFFILIATION)))) $\bowtie$ PLAYER

13

SQL:

```
SELECT PLAYERNAME
FROM PLAYER
WHERE PLAYERNUM NOT IN (
    SELECT PLAYERNUM
    FROM AFFILIATION
)
```

RA: players - not - affiliated =

```
 $\pi$  PLAYERNAME (PLAYER -  $\pi$  PLAYERNAME2
( $\pi$  PLAYERNUM (AFFILIATION  $\bowtie$  PLAYER))
```

14

SQL:

```
SELECT PLAYERNAME, COUNT(DISTINCT TEAMNUM)
FROM PLAYER JOIN AFFILIATION ON
PLAYER.PLAYERNUM = AFFILIATION.PLAYERNUM
GROUP BY PLAYERNAME
```

RA: player - affiliation - count =

```
 $\pi$  PLAYERNAME, COUNT(DISTINCT TEAMNUM)
((PLAYER  $\bowtie$  PLAYERNUM = PLAYERNUM
AFFILIATION)  $\div$  GROUP BY PLAYERNAME
( $\pi$  PLAYERNAME, TEAMNUM (PLAYER)  $\bowtie$ 
PLAYERNUM = PLAYERNUM (AFFILIATION)))
```


SOL:

```
SELECT TEAMNAME, BATTINGAVG  
FROM TEAM JOIN AFFILIATION  
ON TEAM.TEAMNUM = AFFILIATION.TEAMNUM
```

RA: team-batting-average =

```
* TEAMNAME, BATTINGAVG  
(TEAM ⋈ TEAMNUM = TEAMNUM  
AFFILIATION)
```
